In summary, the following things all co-occur during the same abrupt window:

- Capacity for in-context learning sharply improves (as measured via the in-context learning score).
- Induction heads form.
- Loss undergoes a small "bump" (that is, the loss curve undergoes a period of visibly steeper improvement than the parts of the curve before and after).
- The model's trajectory abruptly changes (in the space of per-token losses, as visualized with PCA).

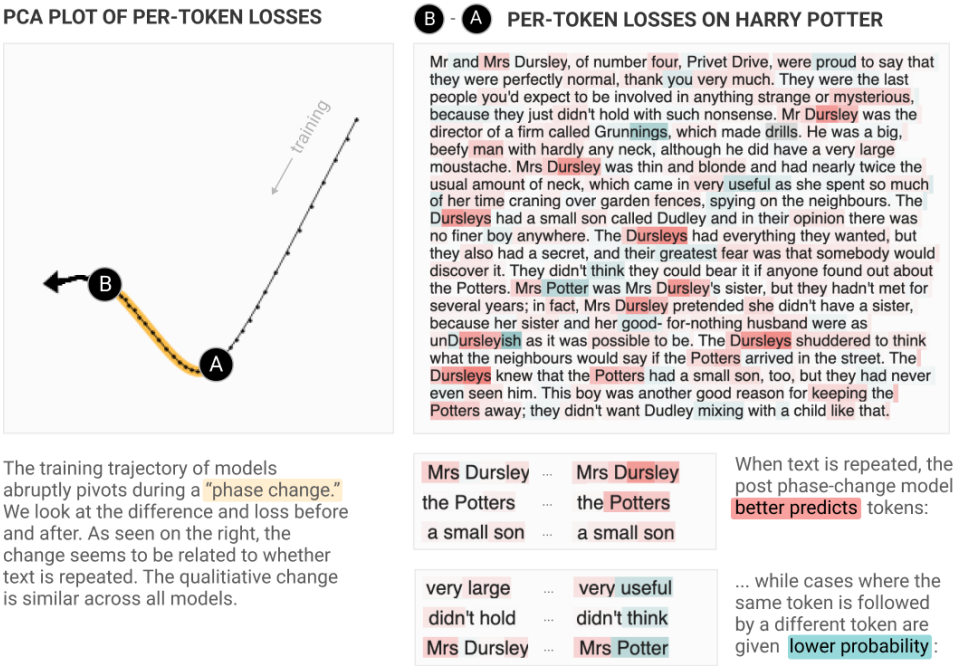
Collectively these results suggest that some important transition is happening during the 2.5e9 to 5e9 token window early in training (for large models this is maybe 1-2% of the way through training). We call this transition "the phase change", in that it's an abrupt change that alters the model's behavior and has both macroscopic (loss and in-context learning curves) and microscopic (induction heads) manifestations, perhaps analogous to e.g. ice melting.[10]

Looking at the Phase Change More Closely

A natural explanation would be that for all these models, the induction heads *implement* in-context learning: their formation is what drives all the other changes observed. To strengthen this hypothesis a bit, we check a few things. First, the window where the phase change happens doesn't appear to correspond to a scheduled change in learning rate, warmup, or weight decay; there is not some known exogenous factor precipitating everything. Second, we tried out training some of the small models on a different dataset, and we observed the phase change develop in the same way (see Model Analysis Table for more details).[11]

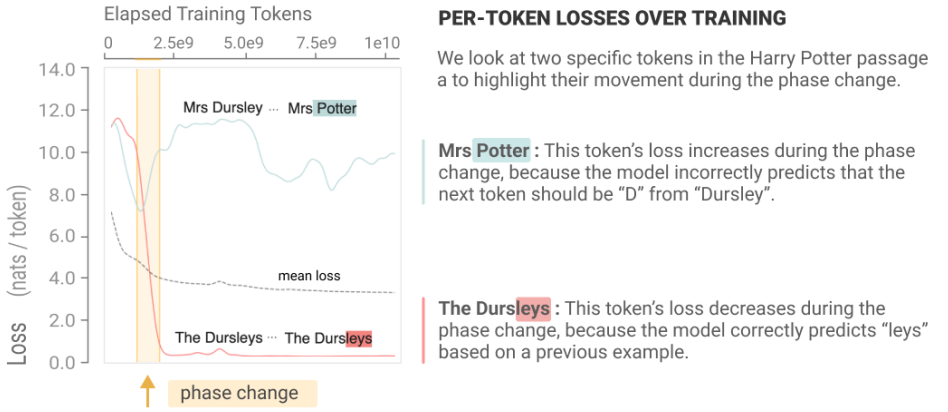
Third, to strengthen the connection a little more, we look qualitatively and anecdotally at what's going on with the model's behavior during the phase change. One way to do this is to look at specific tokens the model gets better and worse at predicting. The model's loss is an average of billions of log-likelihood losses for individual tokens. By pulling them apart, we can get a sense for what's changed.

Concretely, let's pick a piece of text – for fun, we'll use the first paragraph of Harry Potter – and look at the differences in log-likelihoods comparing the start and the end of the phase change.[12] We'll notice that the majority of the changes occur when tokens are repeated multiple times in the text. If a sequence of tokens occurs multiple times, the model is better at predicting the sequence the second time it shows up. On the other hand, if a token is followed by a different token than it previously was, the post-phase-change model is worse at predicting it:



We can also do this same analysis over the course of model training. The loss curve is the average of millions of per-token loss curves. We can break this apart and look at the loss curves for individual tokens.

In particular, let's take a look at per-token loss trajectories for two tokens in the first paragraph of Harry Potter. In red, we show a token whose prediction gets dramatically *better* during the phase change: it's the last of four tokens in " The Durs**leys**", a sequence that appears several times in the text. In blue, we show a token that gets meaningfully *worse* during the phase change: it's the first-ever appearance of " Mrs Potter", after both previous instances of " Mrs" were followed by " Dursley" instead.



All of this shows that during the phase change, we see exactly the behaviors we'd expect to see if induction heads were indeed contributing the majority of in-context learning.

Assessing the Evidence

Despite all of the co-occurrence evidence above (and more in the Model Analysis Table), the fact remains that we haven't shown that induction heads are the primary mechanism for in-context learning. We have merely shown that induction heads form at the same time as in-context learning, and that in-context learning does not improve thereafter. There are a number of potential confounds in this story. Below we summarize reasons in favor and against thinking the connection is causal. The argument in favor might look like this:

- The formation of induction heads is correlated with a great increase in models' capacity for in-context learning, for a wide variety of models large and small.
- It's highly improbable these two sharp transitions would co-occur across so many models by pure chance, without any causal connection.
- There is almost surely some connection, and the simplest possibility is that induction heads are the primary mechanism driving the observed increase in in-context learning. (However, as discussed below, it could also be a confounding variable.)
- Since over 75% of final in-context learning forms in this window, one might naively believe this to be the amount of in-context learning that induction heads are responsible for.

However, the following issues and confounds suggest caution:

- In large models, we have low time resolution on our analysis over training. Co-occurrence when one only has 15 points in time is less surprising and weaker evidence.
- Perhaps other mechanisms form in our models at this point, that contribute to not only induction heads, but also other sources of in-context learning. (For example, perhaps the phase change is really the point at which the model learns how to compose layers through the residual stream, enabling both induction heads and potentially many other mechanisms that also require composition of multiple heads) Put another way, perhaps the co-occurrence is primarily caused by a shared latent variable, rather than direct causality from induction heads to the full observed change in in-context learning.
- The fact that in-context learning score is roughly constant (at 0.4 nats) after the phase change doesn't necessarily mean that the underlying mechanisms of in-context learning are constant after that point. In particular, the metric we use measures a *relative* loss between the token index 500 and 50, and we know that the model's performance at token 50 improves over training time. Reducing the loss a fixed amount from a lower baseline is likely harder, and so may be driven by additional mechanisms as training time goes on.[13] [14]

One point worth noting here is that the argument that induction heads account for most in-context learning *at the transition point* of the phase change is more solid than the argument that they account for most in-context learning *at the end of training* – a lot could be changing during training even as the in-context learning score remains constant.